# UnitedSoft Technologies Inc.

## Digital Attendance & Automated Payroll Mobile System

Official Project Delivery & Completion Status Report

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|---|---|---|---|
| **PROJECT NAME:** | UnitedSoft Mobile Attendance & Automated Payroll | **OVERALL STATUS:** | **100% COMPLETED** |
| **TARGET PLATFORMS:** | Android, iOS (Single React Native Codebase) & Web | **CORE CODEBASE:** | React Native (Expo) |
| **DELIVERY DATE:** | September 27, 2026 | **TEST VERIFICATION:** | **PASSED (0 Errors)** |

The UnitedSoft Mobile Attendance and Automated Payroll System has been architected, developed, and verified end-to-end to streamline enterprise staff management. The system is delivered from a single cross-platform codebase supporting Android and iOS mobile devices along with web preview, backed by a high-performance Express.js REST API with Mongoose MongoDB schemas, JWT authentication, and automated payroll calculations based on real-time attendance telemetry.

| Component | Selected Technology | Purpose / Architectural Role |
|---|---|---|
| Mobile Framework | React Native (Expo SDK 52/57) + TypeScript | Single unified codebase for Android, iOS & Web |
| Backend Engine | Node.js (v24) + Express.js Framework | RESTful API service with CORS, validation & routes |
| Database System | MongoDB (Mongoose Schemas) / Resilient DB | Staff profiles, geofence, leaves, attendance & payroll |
| Security & Auth | JSON Web Tokens (JWT) + bcryptjs hashing | Role-Based Access Control (Admin, HR, Employee) |
| Hardware Telemetry | Expo Location & Expo LocalAuthentication | GPS coordinates, Haversine geofence & biometrics |
| Document Generation | PDFKit Streaming Engine (Server-side) | Official electronic salary vouchers and PDF payslips |
| Data Reporting | Excel-Compatible CSV Export Engine | Company-wide attendance and payroll financial reports |
| API Collection | Postman Collection v2.1.0 | 15+ Pre-configured endpoints with environment testing |

| Module / Feature | Functional Requirement Coverage | Status |
|---|---|---|
| 4.1 Authentication | Email/password login, JWT sessions, 3-tier roles, password reset | 100% VERIFIED |
| 4.2 Staff Records | CRUD employee profiles, salary structure (Base, HRA, PF), search | 100% VERIFIED |
| 4.3 GPS Attendance | Check-In/Out, Haversine geofencing, late detection, overtime, override | 100% VERIFIED |
| 4.4 Leave System | Apply leave, quota tracking (CL 12, SL 10, PL 15), HR approval queue | 100% VERIFIED |
| 4.5 Automated Payroll | 1-Click monthly calculation, Loss of Pay (LOP), PDF payslips | 100% VERIFIED |
| 4.6 HR Dashboard | Live headcount stats, present count, late arrivals, Excel CSV exports | 100% VERIFIED |
| 4.7 Startup Scripts | One-click .bat launchers for backend, database & mobile Expo app | 100% VERIFIED |

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### 4. AUTOMATED PAYROLL COMPUTATION ENGINE FORMULA

The payroll engine automatically executes every month, aggregating attendance records, approved paid leaves, and unexcused absences into an itemized payout statement:

- **Gross Salary = Base Salary + House Rent Allowance (HRA) + Special Allowances + Overtime Pay**
- **Overtime Pay = Overtime Hours * (Base Salary / (Working Days * 8) * 1.5)**
- **Loss of Pay (LOP) = (Unpaid Leave Days + Absent Days) * ((Base Salary + Allowances) / Working Days)**
- **Net Take-Home Salary = Gross Salary - (Loss of Pay + Provident Fund (PF) + Withholding Tax (TDS))**

| File Name | Action & Scope | Target Environment |
|---|---|---|
| start_all.bat | One-Click Master Launcher: boots backend & mobile in parallel | **Windows Cmd/Shell** |
| start_database.bat | Initializes data store, seeds demo records & starts engine | **Node.js Backend** |
| start_backend.bat | Boots Express REST API on http://localhost:5000 | **Port 5000** |
| start_mobile_app.bat | Launches Expo Metro Bundler for Android, iOS & Web preview | **Expo CLI** |

| Role | Email Address | Password \| Designation / Department |
|---|---|---|
| HR Admin | admin@unitedsoft.com | **admin123 \| Sarah Jenkins (Director & System Admin)** |
| Employee | alex@unitedsoft.com | **emp123 \| Alex Morgan (Senior Full Stack Engineer)** |
| Employee | elena@unitedsoft.com | **emp123 \| Brighton Vance (Lead UI/UX Designer)** |
| Employee | david@unitedsoft.com | **emp123 \| David Chen (QA Automation Specialist)** |

### DELIVERY VERIFICATION & ACCEPTANCE CERTIFICATION

This document certifies that the UnitedSoft Attendance & Automated Payroll System has completed comprehensive implementation. All TypeScript modules passed with 0 compile errors, REST API endpoints are functional on port 5000, PDF generation is operational, and .bat startup utilities have been provisioned.